
***EDO STATE COLLEGE OF
AGRICULTURE SCIENCE AND
TECHNOLOGY, IGUORIAKHI,***

***DEPARTMENT OF AGRICULTURAL
TECHNOLOGY***

COURSE CODE: AGT 231

***COURSE TITLE: STATISTICS AND
FIELD EXPERIMENTATION***

COURSE UNIT: 4

S/N	Objectives
1.	Understand simple definitions and concepts statistical data analysis
2.	Understand the sources of statistical data
3.	Understand data presentation
4.	Understand measure of central tendency
5.	Understand measure of absolute dispersion
6.	Understand research methodology
7.	Understand the various statistical tools

STATISTICS

Statistics is the science of collecting, organizing, analyzing, interpreting and presenting data. Statistics involves calculating mathematical probabilities based on data collected from a sample group. The mathematical theories behind statistics rely heavily on differential and integral calculus, linear algebra and probability theory. Statistics are used in virtually all scientific disciplines such as physical and social sciences as well as agriculture, business, medicine, humanities, government and manufacturing.

Importance of Statistics

1. It simplifies mass of data (condensation)
2. It helps in making reliable and objective decision.
3. It presents facts in a precise and definite form
4. It facilitates predictions and forecasting
5. It helps in formulating and testing hypothesis
6. It facilitates comparison

BASIC STATISTICAL TERMS

1. **Population:** A population is the entire set of items, individuals or event that share common characteristics and are the focus of the study. It is the entire group being studied. Population can be finite (measurable and countable) or infinite (theoretical or too large to count).
2. **Sample:** Sample is a smaller manageable subset used to represent the population.
3. **Parameter:** This refers to the numerical characteristics of the entire population.
4. **Censoring:** Censoring is a form of missing data condition in which the value of a measurement or observation is only partially known but a boundary is known. A censored observation provides partial information, allowing for more accurate analysis than ignoring it. It is commonly encountered in survival analysis, reliability engineering and longitudinal studies. Censor can be of right ($T > O$), left ($T < O$), interval ($T_1 < T < T_2$), type 1 and Type 2 (Test ends at fixed time) or random.
5. **Sample survey:** This is the method of collecting data from a subset of individuals or items selected from a larger population to make inferences, estimations or generalizations about the entire group's characteristics. It is designed to be a cost-effective, time efficient and manageable alternative to a full census.
6. **Census:** This is the systematic process of collecting, compiling and publishing data about every single member of a specific population at a particular time, providing a complete snapshot of its characteristics for planning and analysis.

TERMINOLOGIES IN STATISTICS

1. Variable

A variable is a characteristics being measured, counted or observed and whose value can change between data points over time. Variable form the core of data, describing attributes of people, objects of ideas. Variable can be classified based on the type of data and role in research. Choosing which variables to measure is central to a good experimental design.

Based on the type of data, variable can be classified into two: i. Qualitative variable ii. Quantitative variable

1. **Qualitative variable:** This is also called categorical variable and it represents groupings or labels. Qualitative variable can be divided into three.

- a. Nominal variables: These refer to categories with no intrinsic order. Examples include gender, eye colour and marital status.
- b. Ordinal variables: These refer to categories with logical order but the difference between the categories or their averages is meaningless. Examples include pain intensity, education level and satisfaction rating. Ordinal variable is sometimes called ranked variable.
- c. Binary/ Dichotomous variables: These are categorical variables with only two possible categories. Examples include yes/ No, pass/ fail and heads/tails in a coin flip.
2. Quantitative (Numerical) Variables: These represent measurable quantities. It can be discrete or continuous.
 - a. Discrete variables: are countable values often integers e.g. number of children, shoe size etc.
 - b. Continuous variables: are measurable values that can take any value within a range e.g. height, temperature, time, weight etc. Continuous variables can be categorized into two main types based on their scale.
 - i. **Interval variables**: are continuous measurements with a consistent, equal distance between values but they lack a "true zero" point (i.e. zero does not represent the complete absence of the quantity) e.g. temperature in Celsius or Fahrenheit, IQ scores etc.
 - ii. **Ratio**: Ratio variables possess all the features of interval variables but include a "true zero" point, allowing for meaningful comparisons of magnitude (one value can be described as “twice as much” as another). Examples include weight, height, time, distance, Kelvin temperature, etc.

Classification of Variable by Role in Research

1. Independent Variable: Also called outcome variable, is the variable manipulated or controlled to test effects in an experiment.
2. Dependent Variable: This is sometimes known as explanatory, blocking, covariate, or predictor variable. It is the variable being measured or tested.
3. Control Variables: These are variables kept constant during experiment to prevent influencing the results.
4. Extraneous or Confounding Variables: These are uncontrolled variables that might influence the dependent variable, causing false relationships.

2. BLOCK

A block is a group of experimental units that are similar to one another. Blocking is a technique used in the design of experiments to reduce the effects of confounding / uncontrollable variables, leading to a more accurate and reliable results. Hence, blocking is an experimental design technique that arranges subjects or experimental units into homogeneous groups (blocks) before treatment assignment based on one or more variables before to reduce noise and control for known, uncontrollable nuisance variables. The intent of blocking is to prevent large differences between treatments effects, while at the same time, allowing the treatments to be examined under different experimental conditions. Types of blocks or designs include:

- (a) Randomized Complete Block Design (RCBD): This is the most common type where all treatments are randomly assigned within each homogeneous block. Each block acts as a complete replication of the treatments.
- (b) Incomplete Block Design (IBD): This is used when blocks are too small to contain all treatments. It requires specialized analysis to compare treatments.
- (c) Balanced Incomplete Block Design (BIBD): This is a special type of incomplete block design where every pair of treatments appears together the same number of times.
- (d) Matched Pairs Design: This is a specific case of randomized block design where pairs of similar units are matched, and treatments are assigned within each pair to minimize variance.
- (e) Latin Square Design: This is used when there are two blocking factors like rows and columns, in a field. It controls variation in both directions by ensuring each treatment appears once in each row and column.
- (f) Split-Plot Design: This involves nesting of treatments and it is usually used when one factor is harder to change than another. It is useful in agricultural experiments (e.g., irrigation assigned to large plots and fertilizers to subplots).

3. PLOT

A plot is a graphical representation of a dataset, showing distributions, relationships, or trends that are not immediately obvious from raw data. Plots are essential in exploratory data analysis for identifying patterns, verifying statistical assumptions, and visualizing complex information. They transform numerical output into an accessible visual format.

Types of Plots Include:

Box Plot (Whisker Plot): This uses median, quartiles, and minimum/maximum values to summarize data distribution and identify spread and outliers.

Scatter Plot: This displays the relationship between two variables and helps determine correlation.

Histogram: This shows the frequency distribution of a dataset by dividing data into bins.

Mean Plot: This visualizes the means (and sometimes standard deviations) across different groups to detect shifts or changes over time.

Q–Q Plot (or P–P Plot): This assesses how closely two datasets agree or how well a dataset fits a specific model (e.g., normal distribution).

Dot Plot: This is a simple graph used for small samples to represent individual data points and observe frequency.

4. Plot Size

Plot size refers to the area of land assigned to a single experimental unit or treatment. Small plots are preferable when land is limited or non-homogeneous. They allow for more replications, reducing experimental error. Smaller plots often require more buffer zones to manage competition between

treatments. Large plots are necessary to reduce edge effects (impact of adjacent treatments) and to accommodate larger plants, though they may increase land variability. Standard plots are often rectangular in shape (500-900 m²). A typical plot size in Nigeria is 30 m × 30 m (900 m²), Half plot: 15 m × 30 m (450 m²).

Land Measurement Conversions:

1 Acre $\approx$ 6 plots $\approx$ 4046.85 m²

1 Hectare = 10,000 m² ($\approx$ 11 plots)

In estimating plot size, the maximum curvature method is one of the earliest methods used to determine the optimum plot size in field experiments for several crops. However, it may sometimes lead to inaccurate results. Modern approaches use mathematical methods, which are now widely accepted and considered more reliable than visual judgment.

5. CONTROL

Control refers to techniques used to isolate the effect of specific variables by removing the influence of extraneous, confounding, or nuisance factors. It ensures that observed relationships are valid either by holding variables constant in an experiment or by using statistical methods like regression, to adjust for them in data analysis. Control reduces omitted variable bias, increases precision, and ensures validity.

Common Control Methods:

1. Control Variables (Experimental Design): Variables held constant throughout a study to prevent them from affecting the outcome (e.g., keeping room temperature steady).
2. Control Group: This refers to the group that does not receive the treatment in an experiment, allowing for comparison to isolate the treatment's true effect.
3. Statistical Control (Analysis): This involves using mathematical models such as ANCOVA and multiple regression to remove the impact of confounding variables that cannot be physically controlled.
4. Statistical Process Control (SPC): This uses control charts to monitor, analyze, and maintain process stability by minimizing variation.

GERMINATION TEST

A germination test is a standard agricultural procedure used to measure the ability of seeds to germinate and produce normal seedlings under ideal conditions. It is used to determine the percentage of viable seeds in a seed lot and to predict potential field performance by evaluating the number of seeds that produce healthy, normal seedlings. Germination testing is the most common method for evaluating seed quality. It helps to estimate planting value, determine appropriate seeding rates, and assess storage longevity. Germination tests are crucial for identifying seed vigor, ensuring that farmers can predict and achieve optimal planting density, thereby maximizing crop performance.

Purpose of Germination Test

1. It helps in determining seed viability

2. It aids in the calculation of seeding rates
3. It is a quality control measure as it helps in evaluating seed deterioration during storage.
4. It can be used to identify seed-related issues

Methods for Determining Germination Test

Germination test can be carried out using various methods such as Standard warm test (percentage of normal seedlings at ideal temperature 70 °F or 21 °C), Tetrazolium (TZ) (chemical test for quick viability), cold test for early-season vigor, and accelerated aging for storage potential. Common methods include paper towel, top-of-paper and float tests.

1. Float Test (Water Test): This is a simple, quick yet less accurate method where seeds are placed in water. Viable seeds usually sink, while dead / empty seeds float.

2. Ragdoll or paper towel method: This method involves:

- i) Sample selection: Randomly select a representative sample of seeds e.g. 100 seeds.
- ii) Substrate Preparation: Place seeds on a moistened paper towel or between sheets.
- iii) Environment: Place the paper in a sealed plastic bag or container to maintain humidity and keep in a warm (usually 20 - 30 °C), dark place.
- iv) Observation: Check for sprouts after a specific periods (usually 7–14 days).
- v) Calculation: Calculate the percentage germination as follows:

$$\text{Germination (\%)} = \frac{\text{Number of normal seedlings}}{\text{Total number of seeds tested}} \times 100$$

80% and above: Good and acceptable for most crops.

60–70%: Low, may need to increase planting density.

< 50%: Generally considered poor, recommend purchasing fresh, high viability seed.

Evaluation Criteria

1. Normal seedlings: These are seedlings with all essential structures (root, stem, cotyledons) developed enough to produce a plant.
2. Abnormal seedlings: Seedlings that are damaged, deformed or decayed, incapable of normal growth.
3. Dead / Rotten seeds: Seeds that have decayed or did not sprout.
4. Hard / Dormant seeds: Seeds that remain firm but do not germinate, often due to a water-impermeable coat.

LABELING EXPERIMENTS

Labeling experiments in field studies is critical for maintaining organization, reducing experimental error and ensuring data integrity. A proper label includes clear identification of plots, treatments,

replicates and sometimes blinding codes to prevent bias. Based on standard practices, some of the methods for labeling field experiment include:

1. Systematic plot numbering system:

a. Consecutive Numbering: This involves assigning a unique number to every plot in the field e.g. 1-N, where N is the total number of plots.

b. Grid system (row-column): For larger fields, use a coordinate system e.g. row 1 - plot 1, row 1 - plot 2) to easily identify locations.

c. Label stakes: Place a durable stake at the corner of each plot, labeled with the plot number.

2. Label Treatments and Replications: Label each plot clearly with its assigned treatment code e.g., T1, T2, Control etc. Each block or replicate should be labelled clearly to manage field variability e.g., Block I, Block II, etc.

3. Blinding codes: Sometimes, a coding system on plot stakes is used instead of writing the treatment name directly, to avoid biased data collection. For example, a stake can be labelled "plot 001" instead of "100kg Nitrogen" and a separate master list indicating "plot 001 = 100kg Nitrogen" is kept.

Always use waterproof, sun-resistant markers or tags for stakes or bags as the labeling materials.

Sample ID: Always label bags with the date, location, block, plot number and treatment when collecting data.

Example of field plot label:

Trial ID: MK-2026
Block: B3
Plot Number: 12
Treatment: T3

RECORDS

In the process of conducting any experiment, it is crucial to record all the data and observations that are relevant to the experiment. Records in scientific experiments are the systematic documentation of all relevant activities, observations and data, typically maintained in a bound, paginated and signed laboratory notebook or secure digital platform. They serve as the primary, contemporaneous and permanent record of the experimental process and findings, ensuring traceability, reproducibility and compliance with regulatory standards. Components of a good record include experimental details, raw data, calculations or results, interpretations, time stamped and signature, and future steps.

Uses of Records in Experiments

1. It enables reproduction and verification of experimental results.
2. It enables traceability of data & fosters accountability.
3. It serves as legal evidence for the conception of an invention.

4. Through proper record keeping, error can be easily identified and corrected.
5. Record ensures a safe laboratory environment through documentation of equipment calibration, maintenance and usage.

SOURCES OF STATISTICAL DATA

Sources of data collection in statistics can be broadly classified into two: namely primary (first-hand / original) and secondary (previously collected / published) data.

Primary Data Source

Primary data is collected directly by the researcher. Sources of primary data offer high relevance, accuracy, and up-to-date customized insights tailored to specific research goals. It also offers control over data quality and originality. The disadvantage of primary data is that it is time consuming, expensive, may have a limited scope, and there may be potential researcher bias.

Sources of primary data include:

- i) Surveys: This involves the administration of questionnaires in person, via phone, or online to gather specific information from respondents.
- ii) Experiments: Common in scientific research, data are generated from controlled studies to measure the effect of variables.
- iii) Observations: This involves the direct monitoring of behavior or phenomena without direct interaction.
- iv) Interviews: This is a personal, direct conversation to collect detailed qualitative or quantitative data.

Secondary Data Sources

Secondary data are gathered from existing records. They offer significant advantages in cost-effectiveness, time savings, and broad scope. However, there is potential irrelevance to specific research questions, questionable data quality, lack of control over collection methods, risk of being outdated, and risk of incompleteness. Secondary data sources can be categorized into **internal** and **external** data sources.

- i) Internal Data: These are data gathered from within an organization. Examples include HR records and annual profit and loss reports.
- ii) External Data: This refers to information gathered from outside, such as government data or market research from other organizations.

Sources of secondary data include:

- a) Government publication: Data from departments like census bureaus, ministries, finance etc.
- b) Institutional Records: Data collected by organizations for administrative purposes like sales reports or student records.
- c) Academic and published reports: Information from books, journals, internet sources and research.

d) Commercial / private data: Data provided by private agencies or data banks.

SCALES OF MEASUREMENT

Scales of measurement classify data into four hierarchical levels based on their properties. These levels determine the appropriate statistical analyses: Nominal (categories), Ordinal (ranked), Interval (equal distance), and Ratio (absolute zero). The levels range from basic classification to precise quantitative measurement.

1. Nominal scale (lowest level): Data is used for naming or labeling variables without any quantitative value or order. Categories are mutually exclusive and they include gender, eye colour, hair colour, etc.
2. Ordinal scale: Data is arranged in a specific order or rank but the intervals between the ranks are not equal or known. Examples include 1st, 2nd, 3rd place, Likert scales (satisfied, neutral, unsatisfied) etc.
3. Interval: Data is numerical with a consistent / equal distance between points but it lacks a "true zero" point. Addition and subtraction are possible but ratios are not. Examples are temperature in Celsius and IQ scores.
4. Ratio Scale (highest level): Data have ordered, equal intervals and an absolute or "true zero" point. Examples include income, height and weight.

PROBLEMS OF DATA COLLECTION

1. Low response rates often lead to non-representative or biased samples.
2. Adhering to data privacy and compliance regulations is necessary to avoid legal consequences and maintain trust but it is sometimes challenging.
3. Raw data can contain errors, inconsistencies or is incomplete and lead to unreliable insights.
4. Geographical distance and participant reluctance make gathering primary data time-consuming and expensive.
5. Complications may arise from managing the volume, velocity and variety of data from diverse sources.
6. Manual data entry or poorly designed instruments can introduce bias & inaccuracies.
7. Technical difficulty from merging data from multiple fragmented or incompatible sources.

ERRORS OF DATA COLLECTION

Errors in data collection can directly affect the reliability of findings and can lead to misguided conclusions. These errors may be related to the group being studied (sampling errors) or may occur during the actual process of gathering or entering information. These errors include:

1. Sampling errors: These occur when the selected sample does not perfectly represent the entire population. These could be due to selection bias, sample size issues, or sample frame error.

2. Measurement and instrument errors: These occur from the tools or methods used to record data. These may be due to faulty equipment, poorly designed surveys and technical issues like server crashes, software glitches, etc.

3. Human and respondent errors: These are caused by the people giving or collecting the data. These may occur due to respondent bias (giving socially desirable answers instead of honest ones or rushing through a survey due to survey fatigue), non-response errors, and data entry errors.

4. Procedural errors: These occur due to variation in the data collection method as well as inconsistent rounding off of numbers during collection and initial processing stages.

5. Environmental Errors: These occur due to external conditions like loud noise, wind or extreme temperatures that can affect physical measurement or respondent's focus.

DATA PRESENTATION

Data presentation refers to the systematic organization and visual representation of data using tools such as charts, graphs and tables to make complex information clear, actionable and easy to understand. Data presentation is useful in nearly every industry, as it helps professionals share their findings after performing data analysis. The three main methods of data presentation are Textual (descriptive), Tabular (rows and columns) and Diagrammatic/graphical.

Textual Presentation

Data is described in paragraph form, usually used when data is limited and it allows for descriptive interpretation. Textual presentation enables researchers to share information that cannot display on a graph. When a researcher wants to provide additional context or explanation in their presentation, they may choose this format because information may appear clearer in text. Textual presentation is common for sharing research and presenting new ideas. It only includes paragraphs and words, rather than tables or graphs to show data. An example of data you may present textually is findings in a study.

Tabular Presentation

Tabular presentation involves the use of tables to share large amounts of information in rows and columns, depending on the characteristics of the data. It is useful in comparing data and it helps to visualize information. This type of presentation is used in analysis such as:

i) Qualitative classification: Qualities like nationality, age, social status and personal traits may appear in a table to review and compare sociological and psychological information.

ii) Quantitative classification: This category includes items that can be counted or measured.

iii) Spatial/Geographical classification: This applies to situations where information uses a basis of location such as data on a city, state or region.

iv) Temporal classification (chronological): Time is the variable in this category. Any measure of time may help to classify the data.

Advantages of Tabular Presentation

1. It simplifies the data and makes them easily understood to viewers.

2. It helps to provide a side-by-side comparison of the variables you choose.
3. It saves space in presentation because a table condenses the information.

3) Graphical presentation

This uses histograms, lines and curves on graph paper for precise analysis of frequency distributions and time-series data. Graphs provide precise, quantitative and detailed data analysis, and it is preferred for studying functional relationships, trends over time and statistical analysis. Graphs usually represent mathematical relationships (x and y axes) between variables. Types of graphical presentation include:

- i. Histograms: This displays frequency distribution of continuous numerical data using adjacent, equal-width bars.
- ii. Line graphs: This is used to visualize trends and changes in data over a period of time.
- iii. Frequency polygons: Similar to line graphs, these connect the midpoints of histogram bars to show data distribution.

4) Diagrammatical presentation

This involves the use of tools like bar charts and pie diagrams on plain paper for simple, visual comparison of categorical data. Examples include:

- i) Pie charts: This displays any type of numerical data as a fraction in a circle. It works well with fewer variables (2–6). They are simple charts used to depict percentages.
- ii) Bar chart: This uses rectangles of different sizes on an x and y axis to represent different amounts in a dataset. The bars can be plotted vertically or horizontally, and a vertically bar chart is sometimes called a column bar chart.
- iii) Pictograms: These diagrams use images to represent data.
- iv) Cartograms: This includes any type of map that shares the location of a person, place or objects.

MEASURES OF CENTRAL TENDENCY

INTRODUCTION

A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data. As such, measures of central tendency are sometimes called measures of central location. They are also classed as summary statistics. The mean (often called the average) is most likely the measure of central tendency that you are most familiar with, but there are others, such as the median and the mode.

The mean, median and mode are all valid measures of central tendency, but under different conditions, some measures of central tendency become more appropriate to use than others. In the following sections, we will look at the mean, mode and median, and learn how to calculate them and under what conditions they are most appropriate to be used.

MEAN (ARITHMETIC)

The mean (or average) is the most popular and well known measure of central tendency. It can be used with both discrete and continuous data, although its use is most often with continuous data. The mean is equal to the sum of all the values in the data set divided by the number of values in the data set.

When not to use the mean

The mean has one main disadvantage: it is particularly susceptible to the influence of outliers. These are values that are unusual compared to the rest of the data set by being especially small or large in numerical value. For example, consider the wages of staff at a factory below:

Staff	1	2	3	4	5	6	7	8	9	10
Salary	15k	18k	16k	14k	15k	15k	12k	17k	90k	95k

The mean salary for these ten staff is \$30.7k. However, inspecting the raw data suggests that this mean value might not be the best way to accurately reflect the typical salary of a worker, as most workers have salaries in the \$12k to 18k range. The mean is being skewed by the two large salaries. Therefore, in this situation, we would like to have a better measure of central tendency. As we will find out later, taking the median would be a better measure of central tendency in this situation.

Another time when we usually prefer the median over the mean (or mode) is when our data is skewed (i.e., the frequency distribution for our data is skewed). If we consider the normal distribution - as this is the most frequently assessed in statistics - when the data is perfectly normal, the mean, median and mode are identical. Moreover, they all represent the most typical value in the data set. However, as the data becomes skewed the mean loses its ability to provide the best central location for the data because the skewed data is dragging it away from the typical value. However, the median best retains this position and is not as strongly influenced by the skewed values. This is explained in more detail in the skewed distribution section later in this guide.

MEDIAN

The median is the middle score for a set of data that has been arranged in order of magnitude. The median is less affected by outliers and skewed data. In order to calculate the median, suppose we have the data below:

65	55	89	56	35	14	56	55	87	45	92
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We first need to rearrange that data into order of magnitude (smallest first):

14	35	45	55	55	56	56	65	87	89	92
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Our median mark is the middle mark - in this case, 56 (highlighted in bold). It is the middle mark because there are 5 scores before it and 5 scores after it. This works fine when you have an odd number of scores, but what happens when you have an even number of scores? What if you had only 10 scores? Well, you simply have to take the middle two scores and average the result. So, if we look at the example below:

65	55	89	56	35	14	56	55	87	45
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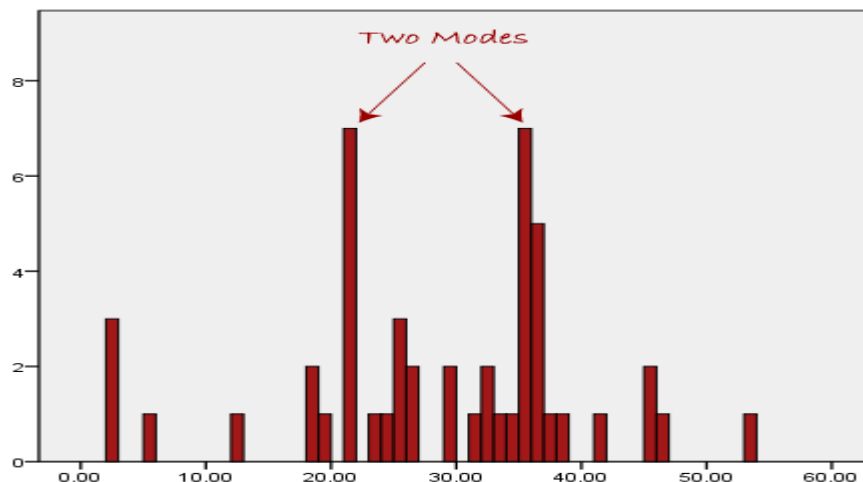
We again rearrange that data into order of magnitude (smallest first):

14	35	45	55	55	56	56	65	87	89
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Only now we have to take the 5th and 6th score in our data set and average them to get a median of 55.5.

MODE

The mode is the most frequent score in our data set. On a histogram it represents the highest bar in a bar chart or histogram. You can, therefore, sometimes consider the mode as being the most popular option. An example of a mode is presented below:



MEASURES OF DISPERSION

Measures of dispersion are non-negative real numbers that help to gauge the spread of data about a central value. These measures help to determine how stretched or squeezed the given data is. There are five most commonly used measures of dispersion. These are range, variance, standard deviation, mean deviation, and quartile deviation.

The most important use of measures of dispersion is that they help to get an understanding of the distribution of data. As the data becomes more diverse, the value of the measure of dispersion increases. In this article, we will learn about measures of dispersion, their types along with examples as well as various important aspects related to these measures.

What is Measure of Dispersion in Statistics?

Measures of dispersion help to describe the variability in data. Dispersion is a statistical term that can be used to describe the extent to which data is scattered. Thus, measures of dispersion are certain types of measures that are used to quantify the dispersion of data.

Measures of Dispersion Definition

Measures of dispersion can be defined as positive real numbers that measure how homogeneous or heterogeneous the given data is. The value of a measure of dispersion will be 0 if the data points in a data set are the same. However, as the variability of the data increases the value of the measures of dispersion also increases.

Measures of Dispersion Example

Suppose we have two data sets $A = \{3, 1, 6, 2\}$ and $B = \{1, 5, 9, 10\}$. The variance(population) of A is 3.5 and the variance(population) of B is 12.68. This implies that data set B is more variable than data set A. Thus, the variance helps to draw a comparison between the two data sets A and B on the basis of variability.

Types of Measures of Dispersion

The measures of dispersion can be classified into two broad categories. These are absolute measures of dispersion and relative measures of dispersion. Range, variance, standard deviation and mean deviation fall under the category of absolute measures of deviation. These measures have the same unit as the data that is being scrutinized. Coefficients of dispersion are relative measures of deviation. Such dispersion measures are always dimensionless. The upcoming sections will further elaborate on these measures.

ABSOLUTE MEASURES OF DISPERSION

If the dispersion of data within an experiment has to be determined then absolute measures of dispersion should be used. These measures usually express variations in a data set with respect to the average of the deviations of the observations. The most commonly used absolute measures of deviation are listed below.

Range: Given a data set, the range can be defined as the difference between the maximum value and the minimum value.

Variance: The average squared deviation from the mean of the given data set is known as the variance. This measure of dispersion checks the spread of the data about the mean.

Standard Deviation: The square root of the variance gives the standard deviation. Thus, the standard deviation also measures the variation of the data about the mean.

Mean Deviation: The mean deviation gives the average of the data's absolute deviation about the central points. These central points could be the mean, median, or mode.

Quartile Deviation: Quartile deviation can be defined as half of the difference between the third quartile and the first quartile in a given data set.

Relative Measures of Dispersion

If the data of separate data sets have different units and need to be compared then relative measures of dispersion are used. The measures are expressed in the form of ratios and percentages thus, making them unitless. Some of the relative measures of dispersion are given below:

Coefficient of Range: It is the ratio of the difference between the highest and lowest value in a data set to the sum of the highest and lowest value.

Coefficient of Variation: It is the ratio of the standard deviation to the mean of the data set. It is expressed in the form of a percentage.

Coefficient of Mean Deviation: This can be defined as the ratio of the mean deviation to the value of the central point from which it is calculated.

Coefficient of Quartile Deviation: It is the ratio of the difference between the third quartile and the first quartile to the sum of the third and first quartiles.

Absolute Measures of Dispersion

Absolute Measures of Dispersion	Formulas
Range	$H - S$ where H is the largest value and S is the smallest value in a data set.
Variance	Population Variance: $\sigma^2 = \sum_1^n \frac{(X_i - \bar{X})^2}{n}$ Sample Variance: s^2 $= \sum_1^n \frac{(X_i - \bar{X})^2}{n-1}$ where n is the number of observations and $\bar{X}$ is the mean
Standard Deviation	Population Standard Deviation: S.D. = $\sqrt{\text{Variance}} = \sigma$ Sample Standard Deviation: S.D. = s

Mean Deviation	$\sum_1^n \frac{ X - \bar{X} }{n}$ where $\bar{X}$ is the central value and denotes the mean, median or mode.
Quartile Deviation	$\frac{Q_3 - Q_1}{2}$ where Q_3 and Q_1 are the third and first quartiles respectively.

Relative Measures of Dispersion

Relative Measures of Dispersion	Formulas
Coefficient of Range	$(H - S) / (H + S)$
Coefficient of Variation	$(S.D. / \text{Mean}) * 100$

Research Methodology

What is Research?

Research as a systematic endeavor to discover, develop, and verify knowledge. The Word 'Research' is comprises of two words = Re+Search. It means to search again. So research means a systematic investigation or activity to gain new knowledge of the already existing facts.

Research is an intellectual activity. It is responsible for bringing to light new knowledge. It is also responsible for correcting the present mistakes, removing existing misconceptions and adding new learning to the existing fund of knowledge. Researches are considered as a combination of those activities which are removed from day to day life.

Research is also considered as the application of scientific method in solving the problems. It is a systematic, formal and intensive process of carrying on the scientific method of analysis. There are many ways of obtaining knowledge. They are intuition, revelation, and authority, logical manipulation of basic assumptions, informed guesses, observation, and reasoning by analogy. One of the branches of research known as empirical research is highly goal-oriented technique.

Characteristics of Research

Characteristics of research determine whether a research is free of biases, prejudices, and subjective errors or not. The terms are very commonly used in research and the success of any research depends on these terms. They can be summarised as:

- i. Generalized.
- ii. Controlled.
- iii. Rigorous.
- iv. Empirical.
- v. Systematic
- vi. Reliability.
- vii. Validity.
- viii. Employs hypothesis
- ix. Analytical & Accuracy.
- x. Credibility.
- xi. Critical

Research Methodology

Research methodology is defined as the systematic study of the techniques applied to identify, select, process, and analyze information about a topic.

Essentially, methodology involves a set of specific procedures or techniques used to identify, select, process, and analyze information about a topic. It guides the research process by addressing two

major questions: what is to be found, and how it is to be found. It is like planning a journey where we first decide where we are going and then we decide how we shall be travelling. We have to identify important stopovers and routes, check points, modes available to reach the destination etc.

Research process

Generally as a process, research is a sequence of activities or steps leading to an end Product. Research process involves a sequence of mutually-inclusive and reinforcing steps and procedures for Acquiring reliable knowledge. There are many diverse ways to describe research process but there are common and essential elements and stages of the process.

- i. Formulating the research problem
- ii. Extensive literature survey
- iii. Developing hypotheses
- iv. Preparing the research design
- v. Determining the sample design
- vi. Collecting the data
- vii. Execution of the project
- viii. Analysis of data
- ix. Hypotheses testing
- x. Generalization and interpretation, and
- xi. Preparation of report and presentation of the results

Component of an Experimental Method Plan

The experimental method is used to establish a cause and effect relationship between variables. The classical experimental study requires three integral components: **manipulation, control, and randomization.**

1. **Manipulation (Intervention/Treatment):** This involves setting the stage for the occurrence of the factor whose performance is to be studied under controlled conditions. The researcher applies a stimulus to a subject and observes the effect.
2. **Control:** This is necessary to hold all other factors constant so that differences between the control group and the treatment group can be linked directly to the applied treatment. Control minimizes the effects of extraneous factors affecting the relationship.
3. **Randomization:** This process ensures that each subject has an equal chance of being assigned to either the experimental group (subjected to treatment) or the control group (not subjected to treatment).
4. **Observation:** The effect of the manipulation of the independent variable on the dependent variable is studied and observed.

Types of Research

- i. **Quantitative research** is based on the measurement of quantity or amount. It is applicable to phenomena that can be expressed in terms of quantity. E.g. Studying the number of enquiries received for room bookings through different modes like internet, emails, calls, letters, or different sources like travel and tours operators, companies and government organizations etc.

- ii. **Qualitative research** is concerned with qualitative phenomenon, i.e., phenomena relating to or involving quality or kind. E.g. studying the stress levels and reasons for variable performances of staff in different shifts in the same department of a hotel. The same individuals may perform differently with the change of shift timings. It can involve performing research about changing preferences of customers as per the change of season.

Quantitative Method of Research

Quantitative research is an approach that deals with **measurable and quantifiable aspects** of the observations. This method focuses on questions such as "to what extent?", "By how much?", and "What relationship exists between factors?"

- **Nature:** It is based on the measurement of quantity or amount and is applicable to phenomena that can be expressed in numerical terms. It involves the collection and analysis of data in numeric form that can be subjected to **rigorous quantitative analysis**.
- **Approach:** The structured approach to inquiry is usually classified as quantitative research. This method typically takes a proactive stance in testing hypotheses and theories.

Characteristics for Designing a Quantitative Survey

Designing a quantitative survey follows a highly structured approach where everything is predetermined.

1. **Predetermination:** All aspects—objectives, design, sample, and the questions to be asked of respondents—must be predetermined.
2. **Rigor and Measurement:** The research uses controlled measurement that is verifiable and replicable.
3. **Focus:** The design should be utilized to determine the extent of a problem, issue, or phenomenon by quantifying the variation (e.g., how many people have a particular attitude).
4. **Systematic Collection:** A quantitative design, such as a sample survey, entails the systematic collection and presentation of data to give a clear picture of a particular situation.
5. **Assessment:** Quantitative surveys typically assess the distribution of established variables and measure their relationships.

Method of Collecting Quantitative Data

Quantitative data collection relies on **controlled measurement** that is verifiable and replicable. Data is typically sought in numeric form.

- **Instruments:** The primary instrument for data collection in quantitative research is often the **questionnaire**.
- **Survey Methods:** Data may be collected through surveys using formal lists of questions (structured surveys), observation, personal interview, telephonic interviews, or schedules.
- **Analysis Focus:** The resulting quantitative data is typically categorized for analysis as univariate (single variable), bivariate (pairs of variables), or multivariate (more than two variables).

Qualitative Method of Research

Qualitative research is concerned with phenomena relating to or involving quality or kind. It seeks to understand the behavior, motivations, desires, attitudes, beliefs, choices, and opinions of people or groups.

- **Nature:** The primary orientation is to describe and analyze culture and behavior and is concerned with understanding the actor's frame of mind.
- **Approach:** It utilizes an unstructured approach to inquiry, allowing flexibility. Qualitative research is typically exploratory, trying to discover (not guess) the conclusion, and aiming to answer the questions "what" and "how".
- **Goal:** It is often aimed at generating bases for theory or hypothesis.

Characteristics for Designing a Qualitative Survey

Qualitative research designs are characterized by flexibility and an exploratory focus.

1. **Flexibility:** The **unstructured approach** allows flexibility in all aspects of the research process, including objectives, design, sample, and questions.
2. **Exploratory Nature:** The design is appropriate for exploring the nature of a problem, issue, or phenomenon without quantifying it.
3. **Descriptive Goal:** The main objective is to **describe the variation** in a phenomenon, situation, or attitude.
4. **Emergent Design:** The design is often **emergent** because the method of research, scope, and types of data collected might change as the research progresses and new insights are revealed.

Method of Collecting Qualitative Data

Qualitative data collection methods involve in-depth interaction aimed at capturing nuanced information. These methods are classified as indirect approaches.

- **Interview Techniques:**
 - **Group Interviews.**
 - **Focus Group Discussion (FGD):** A small group conversation used to gain insight into a population's knowledge, attitude, and behavior.
 - **Key Informant Interview:** Interviews with persons considered experts with special knowledge or insights.
 - **In-depth Interview:** Interviews needed to explore research ideas or hypotheses in detail, using unstructured, free-ranging discussion.
- **Observation:** Techniques such as participant observation.
- **Instrument:** Unstructured interviews generally employ an interview outline called an interview guide.

Experimental Design.

Experimental design, also referred to as the design of experiment, is essentially the logical structure of an experiment. It outlines how observations or measurements must be obtained in order to answer a research question in a way that is valid, efficient, and economical.

The selection of an experimental design depends on your objective (hypothesis + scope). A good experimental design allows the quantification of uncertainty. The experiment is designed so that one can calculate the possibility of obtaining the observed results by chance alone.

Key Objectives and Importance

1. **Ensuring Validity and Precision:** The main objective of experimental design is to provide a mechanism to obtain data so that statistical assumptions are met and analysis can be applied properly. It provides the method by which treatments are randomly placed on experimental units so that responses can be estimated with the utmost precision possible.
2. **Inseparability from Analysis:** The design of the experiment and the subsequent analysis of the data are considered inseparable. Experts stress that the time to think about statistical inference is when planning the experiment. If the experiment is poorly designed, the statistical inferences obtained may be questionable or entirely invalid. As the saying goes, "You cannot save by analysis what you bungle by design".

Core Components

The logical structure of a good experiment is defined by several key components:

- **Treatment structure:** This defines the treatments used and how they relate to one another.
- **Design structure:** This determines how treatments are assigned to the experimental units.
- **Replication:** This is the specific number of experimental units that will be subjected to each treatment.
- **Response structure:** This covers the set of response variables to be measured and the plan for sampling those variables.
- **Error control:** This involves using strategies, such as blocking techniques (local control) or standardized procedures, to reduce unwanted variation or "noise".

There are three basic statistical requirements for building a good experiment: **Randomization, Replication, and Local Control.**

1. Randomization

Randomization is a fundamental principle of experimental design that focuses on the assignment of treatments to experimental units.

Randomization is the statistical process used to ensure that each treatment has an equal chance of being assigned to any experimental unit. This process helps to make observations random.

In a completely randomized design (CRD), randomization means that treatments are assigned completely at random so that every experimental unit has an equal chance of receiving any one treatment. If the randomization process ensures that every experimental unit has an equal chance of receiving each treatment, it is called **complete randomization**.

The general procedure for randomization involves:

1. Determining the total number of experimental units.
2. Assigning a plot number consecutively from 1 to each experimental unit.
3. Assigning the treatments to the units using a chosen randomization scheme, such as a table of random numbers, a random number generator, or drawing lots.

Functions and Importance

The core purpose of randomization is to protect the validity of the experiment and enable proper statistical inference. Its primary functions include:

1. **Neutralizing Systematic Bias:** Randomization helps **eliminate systematic bias** in the experiment. Systematic bias occurs when extraneous, unknown factors might unfairly favor one treatment over another. For instance, if one variety of maize is unintentionally placed only in the most fertile area of a field, the comparison would be biased; proper randomization prevents this.
2. **Ensuring Independence of Errors:** Randomization is crucial because it ensures the **independence of errors** (the random part of the variation), which is an essential assumption for many statistical analyses, such as the analysis of variance (ANOVA).
3. **Distributing Variation:** It helps in distributing unknown variation caused by "confounded variables" throughout the entire experiment, effectively breaking their influence.
4. **Obtaining a Representative Sample:** Randomization is needed to help obtain a **representative sample** from the population being studied.

Randomization is considered one of the **most important basic principles** in designing experiments, alongside replication. It forms the basis of a valid experiment, though replication is also necessary for validity.

2. Replication

Replication is one of the three basic statistical principles essential for a good experimental design, along with randomization and local control.

Definition of Replication

Replication refers to the number of experimental units to which a given treatment is applied. It is the process of repeating the experimental situation by applying each treatment to more than one experimental unit.

The total number of experimental units used in an experiment (N) is equal to the number of treatments (t) multiplied by the number of replications (r) ($N = r \times t$). The experimental unit is the

smallest item to which a treatment is assigned and on which an observation or measurement is made, such as a field plot, an animal, or a cage.

Functions of Replication

Replication is fundamental because it performs critical functions necessary for a valid and precise experiment:

Measuring Experimental Error: Replication is the only way to obtain a measure of the experimental error. Experimental error is the inherent variation that exists among experimental units that are treated exactly alike. This measure of error is necessary for deciding whether an observed difference between treatments is real or simply due to chance.

Improving Precision: Replication provides an efficient way of increasing the precision of an experiment. Precision measures how close the observed values are to each other. When you increase the number of replicates (n) for a treatment, the standard error for the estimate of the treatment effect is lowered, thereby making the experiment more precise. The variance of the sample mean decreases as the number of observations (n) increases.

Increasing the Scope of Inference: Replication also helps to increase the intended scope of conclusions that can be drawn from the experiment.

Replication vs. Repeated Measurement/Subsampling

It is crucial to correctly identify what constitutes a replication:

Replication involves independent experimental units receiving the same treatment.

Repeated Measurements on the same experimental unit (e.g., measuring yield of a perennial crop on the same plot every year) are typically not considered true replications.

Subsampling (taking multiple measurements within the same experimental unit, like dividing one plot into three subplots and measuring each) can yield a more precise value for that specific unit, but the subplots themselves are not replications because the treatment was only applied once to the larger unit. To determine what counts as a true replication, you must identify what was randomized—the unit to which the treatment was randomly assigned is the experimental unit, and the number of these independent units per treatment is the number of replications.

Replication is considered, along with randomization, one of the most important basic principles in designing experiments, and it is necessary for the validity of the experiment.

3. Local Control or Blocking

Local control, often implemented through blocking, is one of the three basic statistical requirements for designing a good experiment, alongside randomization and replication. Local control is a technique adopted in experimental design primarily to reduce or minimize experimental error. You use this principle primarily when your experimental material or units are not homogeneous.

The process of local control, or blocking, involves:

1. **Grouping similar units:** Experimental units that are alike or relatively homogeneous are grouped together into sets called **blocks**.
2. **Assigning treatments:** All treatments are then assigned to units within each block separately and independently.
3. **Removing variation:** This strategic grouping allows for the measurement of variation that exists *among* the blocks, which is then mathematically removed from the experimental error.

For example, if you are conducting a field trial where the soil fertility gradually changes across the area, you would create blocks (rows or columns) that are internally homogeneous, and then randomize all treatments within each block. This removes the "blocking effect" (the fertility difference) from your calculation of experimental error.

Purpose and Impact

The primary goal of employing local control is error control or "noise" reduction.

- **Reduction of Error:** By eliminating known sources of variability (the "blocking effect"), the overall experimental error is reduced.
- **Increased Efficiency and Precision:** When the experimental error is decreased, the experiment's ability to detect existing or real differences among the treatments being tested is increased. This improves the efficiency and precision of the experiment.
- **When to Use:** Local control is necessary when the experimental material is not homogeneous. For instance, in a field experiment, if the soil fertility changes gradually across the area, defining blocks of similar fertility ensures that the difference in fertility (the unwanted variability) is controlled and removed from the experimental error.

Designs Utilizing Local Control

Local control is utilized in various experimental designs depending on the number of non-homogeneous factors present:

- **Randomized Complete Block Design (RCBD):** This design utilizes local control to eliminate one known source of variability (the difference among blocks) by grouping units into homogeneous blocks.
- **Latin Square Design (LSD):** This design handles two known sources of variation simultaneously by grouping units into both rows and columns (row blocking and column blocking), which are estimated and removed from the experimental error.
- **Graeco-Latin Square Design (GLSD):** This design provides triple blocking variation, allowing for the elimination of three sources of variation, thus offering even greater potential for error reduction through planning.